

Field Isolation

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There are many ways to isolate an area of the mouth or a tooth so that restorative services can be performed without interference from soft tissues, the tongue, saliva, or other fluids. Various tongue- and cheek-retracting devices and suction methods may be used; some of these are discussed later in this chapter. By far the most complete method of obtaining field isolation is the rubber dam, the primary subject of this chapter.

Rubber Dam

Sanford C. Barnum is credited with introducing the rubber dam to the dental profession in 1864.[1] For many years, the rubber dam has been recognized as an effective method of obtaining field isolation, improving visualization, protecting the patient, and improving the quality of operative dentistry services. It has been demonstrated that most patients prefer the use of the rubber dam for restorative procedures.[2–4] The dam has been acknowledged as an important barrier for prevention of microbial transmission from patients to members of the dental care team. In addition, it is medicolegally prudent to use a dam for procedures in which small objects, such as dental burs or endodontic files, could be aspirated by the

patient.

Experts in restorative dentistry[5,6] have emphatically stated that the use of the rubber dam not only boosts the quality of restorations but also increases quantity of restorative services because patients are unable to talk or expectorate when the dam is in place. They have further stated that the operating field can only be maintained free of saliva and other contaminants with the dam in place, and the field is more accessible, airborne debris is reduced, and the patient feels more comfortable.

Complete isolation is important for the operative field, but not specifically the use of the rubber dam. One study[7] evaluated longevity and performance of facial cervical resin composite restorations: two-thirds were performed with rubber dam isolation and one-third with thorough isolation using a saliva ejector, cotton rolls, and gingival retraction cord. In 3 years, there was no significant difference in performance of the restorations between the two groups, and more than 95% of the restorations were completely retained; of those placed using non–rubber dam isolation methods, all were retained at 3 years.

There is some evidence that indicates that rubber dam use during resin bonding procedures involving enamel leads to improved bond strengths[8] and reduced microleakage,[9] while other studies have shown no difference in outcomes for restorations placed with rubber dam or good cotton roll isolation (in cavities with enamel margins).[10–14] There is also evidence that salivary contamination can adversely affect the bond strength provided by some dentin bonding systems[15–17] and that restorations placed with a rubber dam in place prior to cavity preparation provided higher bond strength than if the rubber dam was placed after preparation.[18] It should again be emphasized that avoidance of contamination of the operating field is what is important in bonding procedures and other operative procedures and that adequate isolation can often be obtained by measures other than the use of the dam.

Most dentists are taught the use of the rubber dam in dental school, and many suffer tremendous frustrations during rubber dam applications. For the dam to be used and to actually save chair time, the practitioner must be able to apply it quickly and easily. This chapter is designed to describe methods that facilitate use of the rubber dam.

Instruments and materials

Rubber dam material

Rubber dam materials are currently available in an array of colors, ranging from green to lavender to gray to ivory. It is important in operative dentistry to use a dam color that contrasts with the color of teeth; the ivory-colored dam is therefore not recommended for operative dentistry procedures. The original gray dam is still available, but the bright colors have gained popularity. Some operators use the gray dam because they believe that it is better for matching shades in tooth-colored restorations. Because shades of restorative materials are selected prior to rubber dam placement and tooth color changes with the enamel desiccation that accompanies rubber dam use, the restorative shade is probably not affected by the use of a brightly colored rubber dam.

Because of the increased incidence of latex allergies, a number of nonlatex dams are now available. Some dental practices have gone to nonlatex dams exclusively as part of their efforts to be latex free. More information on latex allergies appears later in this chapter.

Rubber dam material is available in rolls, either 5 or 6 inches wide, from which squares may be cut. It is also available in sheets that are 5 inches square, usually used for children, and 6 inches square.

Rubber dam material is available in several thicknesses, or gauges (Table 8-1). The heavy and extra-heavy gauges are recommended for isolation in operative dentistry. If the rubber of the heavier gauges is passed through the interproximal tooth contacts in a single thickness and not bunched in the contacts, the heavy dams are no more difficult to apply than are the thinner materials, and heavier dams are less likely to tear. The heavier materials provide a better seal to teeth and retract tissues more effectively than the thinner materials.

Table 8-1 **Available rubber dam thicknesses (gauges)***

Gauge	Thickness (range)*
Thin	0.006 (0.005—0.007) inch
Medium	0.008 (0.007—0.009) inch
Heavy	0.010 (0.009—0.015) inch
Extra heavy	0.012 (0.0115—0.0135) inch
Special heavy	0.014 (0.0135—0.0155) inch

*Thickness ranges listed by Hygenic.

Latex rubber dam material has a shelf life of more than a year, but aging is accelerated by heat. Extra boxes of dam material can be stored in a refrigerator to extend the shelf life. Latex dam material that has exceeded its shelf life becomes brittle and tears easily; unfortunately, this is usually noticed during dam application. A simple test for the resistance of rubber dam material to tearing is to attempt to tear a sheet grasped with thumbs and index fingers; a strong dam will be very difficult to tear. Brittle dam material should be discarded. If the material was recently purchased, it should be returned to the supplier for replacement.

Napkin

The rubber dam napkin is a piece of strong, absorbent cloth or paper placed between the rubber dam and the patient's face. The napkin provides greater comfort for the patient, especially during unusually long procedures. Napkins are available in two shapes (Fig 8-1). The smaller napkin is usually used with rubber dam frames; the larger provides padding for the side of the face when retracting straps are used.



Fig 8-1 Rubber dam napkins (Hygenic) for longer procedures. Napkins provide padding between the rubber dam and the face and lips, making the dam more comfortable for the patient. The small napkin is for use with rubber dam frames. The larger napkin is for use with strap- or harness-type rubber dam holders.

Punch

At least two types of rubber dam punches are available (Fig 8-2). The Ainsworth-type punch, which is made by several manufacturers, is excellent if it is well made. The Ivory punch (Heraeus Kulzer) is also excellent and has a self-centering coned piston, or punch point, which helps to prevent partially punched holes (Fig 8-3). Punches should have hardened steel cutting tables (or anvils) with a range of hole sizes so that the dam will seal against teeth of various cervical dimensions (Fig 8-4).

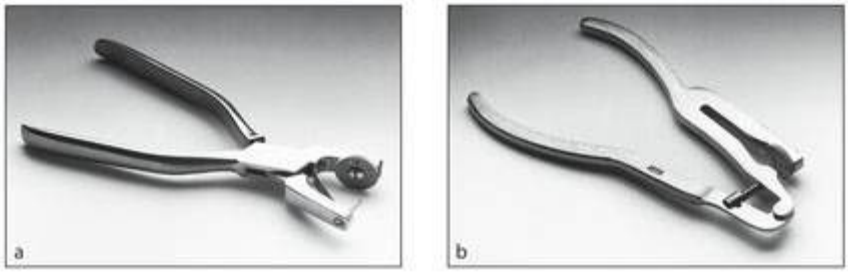


Fig 8-2*(a)* Ainsworth-design (Hygenic) rubber dam punch. *(b)* Ivory-design rubber dam punch.

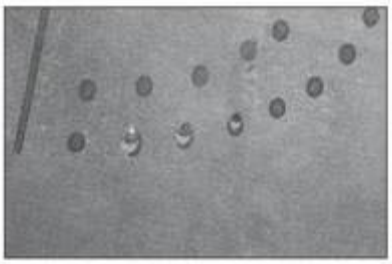


Fig 8-3 Partially punched holes. Stretched rubber dam shows the flaps of dam material left when holes are incompletely punched. The flaps will prevent proper seal. If the flaps are torn off, ragged edges can lead to tearing of the dam during application.



Fig 8-4 The cutting table, or anvil, of a rubber dam punch should have a range of hole sizes. Pictured is the cutting table from an Ivory punch.

Occasionally, the rim of a hole may be damaged because the rotating cutting table was not snapped completely into position before an attempt was made to punch a dam. Holes must be cleanly cut; incompletely punched holes (see Fig 8-3) will promote tearing of the dam during application or will affect the ability of the dam to seal.

A damaged hole rim in the cutting table will cause incomplete cutting. A damaged wheel should usually be replaced; the manufacturer of the punch can replace a damaged wheel.

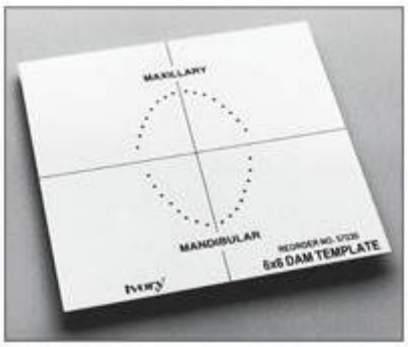
Hole-positioning guides

Although many operators punch the holes without a positioning aid, most find it

helpful to have some form of guide to determine where the holes should be punched. There are several ways to mark a rubber dam so that holes can be located optimally.

Teeth as a guide. The teeth themselves, or a stone cast of the teeth, can be used in marking the dam. To use this method, the dam is held in the desired position in the mouth, or on the stone cast, over the teeth to be included in the isolation. The cusp tips of posterior teeth and incisal edges of anterior teeth can be visualized through the dam, and the centers of the teeth are marked on the dam with a pen. An advantage of this method is precise positioning of the marks even when teeth are malaligned. Its disadvantages include the time-consuming nature of the procedure and the inability to punch a dam before the patient is seated.

Template. Templates are available to guide the marking of the dam (Fig 8-5). These templates are approximately the same size and shape as the unstretched rubber dam itself. Holes in each template correspond to tooth positions. The template is laid over the dam, and a pen is used to mark through selected holes onto the dam. With the template, the dam can be marked and punched before the patient is seated.



Ivory template for marking the dam. Marks corresponding to the teeth to be isolated are made on a 6-inch rubber dam through the holes with a felt-tipped or ballpoint pen.

Rubber dam stamp. Rubber stamps provide a very convenient and efficient way of marking the dam for punching (Fig 8-6). There are commercially available stamps, or stamps can be made by any rubber stamp manufacturer from a pattern, such as the one shown in Fig 8-7, or any custom design. Dams should be prestamped by an assistant so that the marks for the maxillary central incisors are positioned approximately an inch from the top of the dam. Exceptions to normal tooth position are easily accommodated.



Fig 8-6 Rubber dam stamp for the adult dentition.

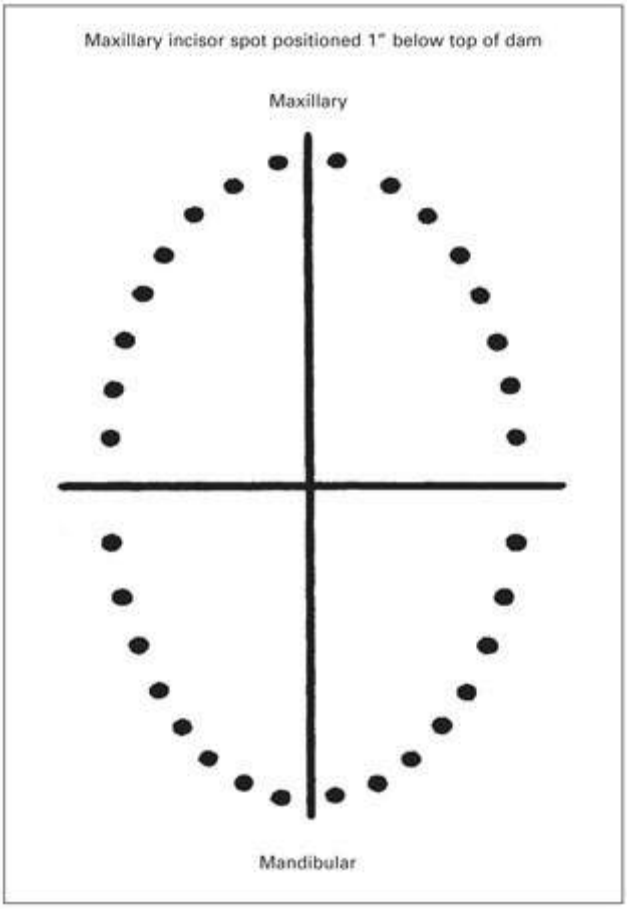


Fig 8-7 Pattern for a rubber dam stamp. This may be duplicated and taken to a rubber stamp manufacturer.

Rubber dam holders

Strap holders. Strap holders (Fig 8-8) provide the most cheek and lip retraction, access, and stability but may cause the most discomfort to the patient. A rubber dam napkin is a necessity for patient comfort when a strap holder is used. The Woodbury retractor grasps the dam material with spring-loaded clips. When posterior teeth are

isolated with a Woodbury-type holder, a tuck or fold in the dam may be needed (Fig 8-9).

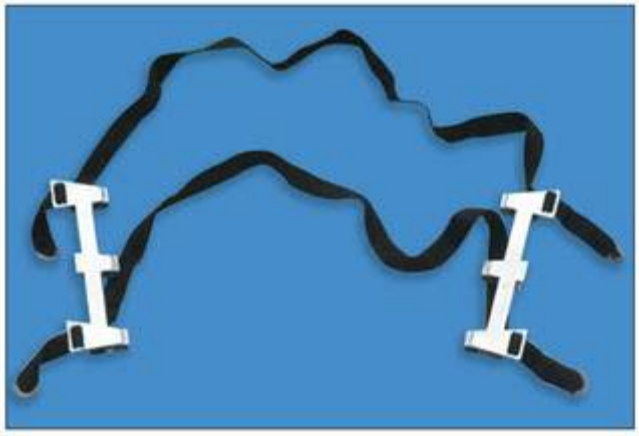


Fig 8-8 Strap- or harness-type rubber dam holders provide excellent lip and cheek retraction. Pictured is a Woodbury retractor (Sulter Dental).



Fig 8-9 A fold or tuck is made in the rubber dam to provide an uncluttered operating field.

Frame holders. Frame holders are exemplified by the Young frame (Young Dental) and the Nygaard-Ostby frame (Coltène/ Whaledent) (Fig 8-10). A U-shaped Young-type frame is made by several manufacturers in both metal and plastic. The Young-type frames are available in both adult and child sizes. A plastic frame is advantageous when radiographs will be a part of the procedure because it is radiolucent. The plastic frames do not, however, stand up to heat sterilization as well as do metal frames, and they have a shorter life span. Metal frames are less bulky and last for years.[19] They are available with balls on the ends to protect the patient in the event that the frame is inadvertently pushed toward the eyes.

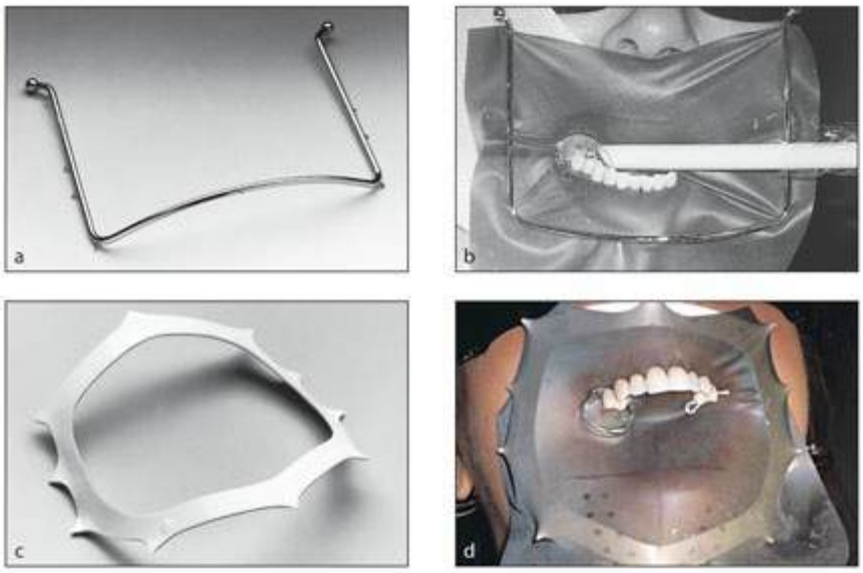


Fig 8-10*(a)* Metal Young frame with eye protectors. *(b)* Young frame inserted into the external surface of the dam. *(c)* Plastic Nygaard-Ostby frame. *(d)* Nygaard-Ostby frame inserted into the internal surface of the dam.

The Young frame is usually positioned on the outside surface of the dam so that it is not in contact with the patient's face. The Nygaard-Ostby frame is normally positioned on the tissue surface or inside surface of the dam and touches the patient's face (or the rubber dam napkin). All frames have points or pegs over which the dam material is stretched to provide a clear operating field and to hold the frame in position. Some Young-type frames come with a hook on each side for attachment of a strap. The strap is run around the back of the head and can be tightened to pull the frame posteriorly to better retract lips and cheeks. If the operator doesn't find the strap useful, the hooks may be cut off, leaving an additional point on each side of the frame for attachment of the dam.

Preattached frames. One commercially available rubber dam (HandiDam, Aseptico) comes with a built-in frame and a rod for insertion to keep the dam open (Fig 8-11). OptraDam Plus (Ivoclar Vivadent) incorporates an integral frame that fits intraorally to assist in cheek retraction. This device may often be used without clamping. A similar device, OptraGate (Ivoclar Vivadent) is useful for retracting lips and cheeks in the anterior region. While not strictly a field isolation device, it helps to improve access and visualization of the field.



Fig 8-11 In the HandiDam, the frame is an integral part of the dam.

Clamp forceps

Ivory-type clamp forceps (Fig 8-12a) are available from several manufacturers and with differently angled beaks. Ivory forceps (Heraeus Kulzer) have stabilizers that prevent the clamp from rotating on the beaks (Fig 8-12b). This is usually advantageous, but it limits the use of these forceps to teeth that are within a range of normal angulation.

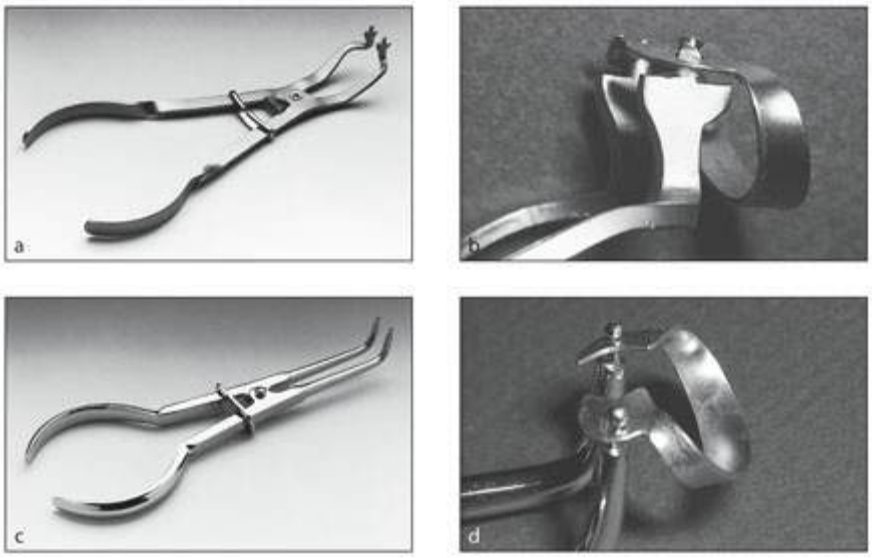


Fig 8-12*(a)* Ivory forceps. *(b)* Stabilizers near the tips of the Ivory-type forceps limit rotation of the clamp when it is held by the forceps. *(c)* Stokes-type forceps. *(d)* The tip design of the Stokes forceps provides more freedom for rotation of the clamp while it is held by the forceps.

Stokes-type clamp forceps (Fig 8-12c), which have notches near the tips of their beaks to locate the holes of a rubber dam clamp (Fig 8-12d), allow a range of rotation for the clamp so that it may be positioned on teeth that are mesially or distally angled.

Either of these types of clamp forceps will serve the practitioner well, and selection should be based on personal preference. The Ivory-type forceps are probably the most popular because of cost.

Clamps

Rubber dam clamps are the usual means of retaining the rubber dam. The three basic types of clamps and their parts are shown in Fig 8-13. When a posterior segment is isolated, the clamp is usually placed on the distalmost exposed tooth (Fig 8-14). The clamp may also be placed on an unexposed tooth (one for which a hole has not been punched) (Fig 8-15).

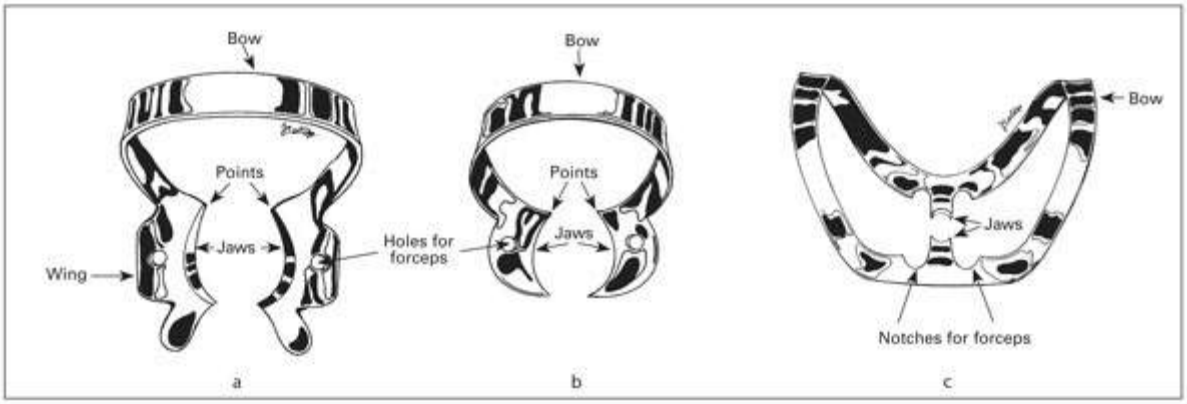


Fig 8-13*(a)* Winged rubber dam clamp; *(b)* wingless rubber dam clamp; *(c)* butterfly rubber dam clamp.



Fig 8-14 Isolated mandibular right quadrant. The clamp is positioned on the distalmost exposed tooth.



Fig 8-15 Isolated mandibular left quadrant with a second clamp placed on an unexposed molar on the right side of the mouth to give additional access to the lingual surfaces of the teeth in the left quadrant. The dam has been loosely stretched over the unexposed tooth to prevent the clamp from initiating a tear. The mirror and other instruments will now be unimpeded when the defective lingual margin of the crown on the first molar is treated.

There are clamps with jaw sizes to fit every tooth. Some clamps simply have a number designation; others have a W in front of the number. The W indicates that the clamp is wingless (see Fig 8-13b); those clamps that do not bear a W have wings (see Fig 8-13a) so that the dam may be attached to the wings before the clamp is placed on the tooth (Fig 8-16).



Fig 8-16 Winged clamp attached to the rubber dam. The edges of the hole are stretched over the wings of the clamp.

Manufacturers have reduced the number of clamps they produce, but a variety of clamp designs remain available. For the practice of operative dentistry, the number of clamps should be limited to a few that will satisfy most needs; these may be kept in the instrument kit and sterilized along with the other operative dentistry instruments. Clamps that will serve in most situations and are recommended for inclusion in operative dentistry instrument kits are listed in Table 8-2 and shown in Fig 8-17.



Fig 8-17*(left)* Clamps recommended for routine use: *(top row, left to right)* no. W8A, B1, 27; *(bottom row)* no. W2A clamp and no. 212SA retractor.

Table 8-2	**Clamps recommended for inclusion in operative dentistry instrument kits***		
Wingless clamps	**Winged clamps**	**Tooth fit**	**Note**
W8A, W8ASA,† or B1	8A or 27†	Molar	Bow extended distally in 27
W2A†	2A	Premolar	

212SA†		Premolar, canine, and incisor	For Class 5 isolation

*All clamps except B1 (Hygenic) and W8ASA (Hu-Friedy) are from Ivory catalog(Heraeus Kulzer).
†Recommended to be available for routine use.

Supplemental clamps, to be available on the rare occasions when the usual clamps will not suffice, should be packaged and sterilized separately. Recommended supplemental clamps are listed in Table 8-3 and shown in Fig 8-18.



Fig 8-18*(right)* Recommended supplemental clamps: *(left to right)* no. 00 (for mandibular incisors and other small teeth), no. W14A (for partially erupted molars), and no. W1A (for premolars with subgingival margins).

Table 8-3	**Recommended supplemental clamps***		
Wingless clamps	**Winged clamps**	**Tooth fit**	**Note**
W0	00†	Small incisor	
W1A†	1A	Premolar	Gingivally angled jaws
W14A†	14A†	Molar	For partially erupted molar

*All clamps are from Ivory catalog (Heraeus Kulzer).
† Clamps recommended to be available to supplement clamps listed in Table 8-2

No. W8A clamp. Although Ivory modified the design of the no. W8A clamp many years ago so that the jaw points do not extend so severely in a gingival direction, some still extend further gingivally than is desirable. The jaws of a no. W8A clamp, for most applications, should be approximately horizontal (Fig 8-19a, bottom) prior to expansion of the clamp for placement on a tooth. As the jaws are spread, the angle of the jaws will change to a gingival orientation; this is usually desirable, but before

the clamp is expanded, the jaws should have little or no gingival angulation.



Fig 8-19 Modification recommended for no. W8A clamps, to thin the jaws and reduce the extension of the jaw points toward gingival tissue. *(a) (top)* Clamp as received from the manufacturer; *(bottom)* clamp that has been modified. *(b)* The points are trimmed from the tissue side so that gingival extension is reduced and jaws are thinned. The bur being used in the high-speed handpiece is a no. 7803 bullet-shaped finishing bur. *(c)* Points that have been sharpened during modification must be dulled. If the points are left sharp, they can damage the surface of the tooth.

For no. W8A clamps in which the jaws have a significant gingival angulation, a modification procedure is recommended (Fig 8-19) unless deep subgingival placement of the points is needed. This modification may be made with a stone used in a low-speed handpiece or a finishing bur used in a high-speed handpiece. After the modification is made, the points, which have been sharpened by the modification procedure, must be blunted to prevent damage to tooth surfaces.

The no. W8ASA clamp (Hu-Friedy) incorporates most of the advantages of the modification of the no. W8A (Fig 8-20).

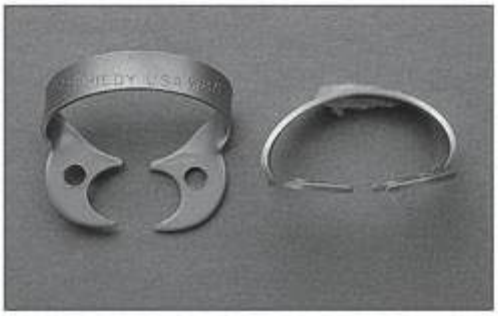


Fig 8-20 No. W8ASA clamp featuring most of the benefits of the modified no. W8A clamp shown in Fig 8-19.

Butterfly clamps. Most of the clamps listed in Tables 8-2 and 8-3 may act as *rubber dam retainers* (placed on the distal tooth or teeth to hold the dam on the quadrant or arch) or as *rubber dam and gingival tissue retractors* (to retract the dam and tissues away from a preparation margin in the cervical area of a tooth). One clamp, however, the butterfly clamp, no. 212SA (Fig 8-21a), is designed to serve as

a retractor only. Because of its double bow and the closeness of the points of each jaw, this clamp must often be stabilized on the tooth (Fig 8-21b), or it may rock mesiodistally during the procedure and damage the root. For retraction for a facial Class 5 restoration, dental impression compound (such as red or green compound, Kerr/ Sybron) should be used under the bows of the clamp on the occlusal (or incisal) and lingual aspects of the teeth to provide stabilization. The hole for the tooth that is to receive the no. 212SA retracting clamp should be offset facially from the line of teeth by approximately 1 mm (Fig 8-21c). This provides for a greater width of interdental rubber dam septum so that, when the dam is displaced apically for access to the gingival margin of a Class 5 restoration, a water-tight seal will be maintained.

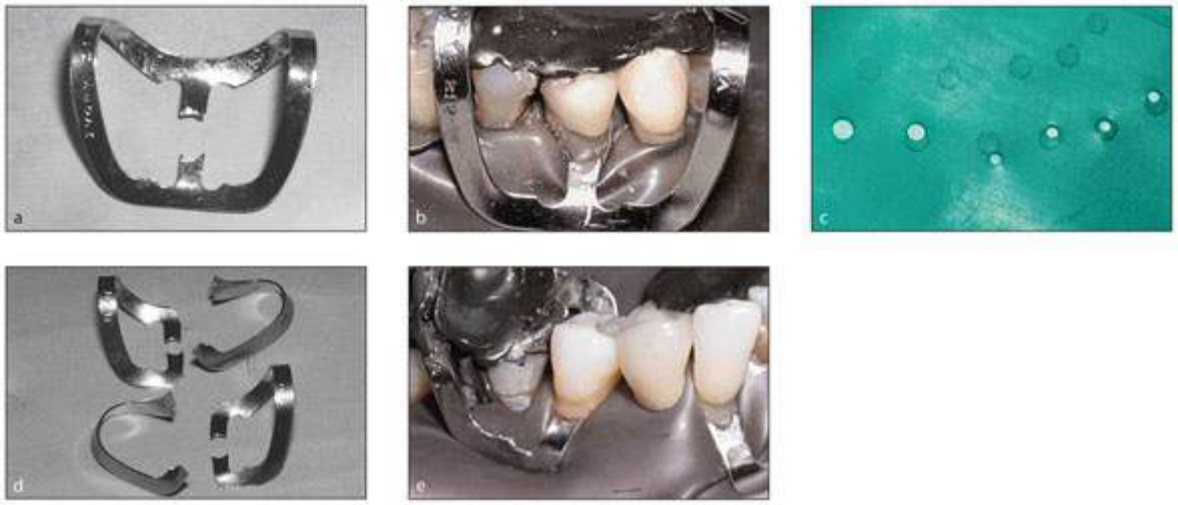


Fig 8-21*(a and b)* No. 212SA clamp (or retractor) for retracting the gingival tissue and rubber dam. *(c)* When a no. 212SA retracting clamp is to be used to retract the dam for a facial Class 5 restoration, the hole for the tooth to be restored should be punched approximately 1 mm facial to the normal hole location. *(d and e)* Two no. 212SA retractors may be modified to give two clamps for tissue and dam retraction for side-by-side restorations. The no. 212SA and the modified no. 212SA retractors must be stabilized with modeling compound or the jaws will damage tooth surfaces.

The double bow of the no. 212SA clamp precludes placement of two clamps on adjacent teeth. When two Class 5 restorations are to be placed on adjacent teeth, two no. 212SA clamps may be modified (Fig 8-21d); one of the bows of each clamp is cut off so that the remaining bow of one clamp extends to the right and the bow of the other extends to the left. If these clamps are stabilized with modeling compound, adjacent Class 5 restorations may be accomplished simultaneously (Fig 8-21e). A no. 212SA clamp or a modified no. 212SA clamp may be used on one root of a molar that has a long clinical crown as well as on single-rooted teeth.

Tooth contact. An important consideration when a clamp is selected is that only its jaw points contact the tooth; this gives four-point contact (Fig 8-22). No clamp jaw can ever be contoured to fit a tooth precisely, nor is there any reason for a clamp to fit precisely, because the dam, not the clamp, creates the seal. Molar clamps should have accentuated arches between the jaw points to ensure that the points are in contact with the tooth, even in teeth with very convex cervical areas. The distance between the points of a jaw, along with the stiffness of the bow of the clamp, determines the stability of the clamp. If there is contact between the tooth and any other part of the clamp's jaw, the contact points are brought closer together, thus reducing the stability of the clamp and allowing it to rotate on the tooth and, occasionally, to be dislodged from the tooth. Four-point contact is therefore very desirable.

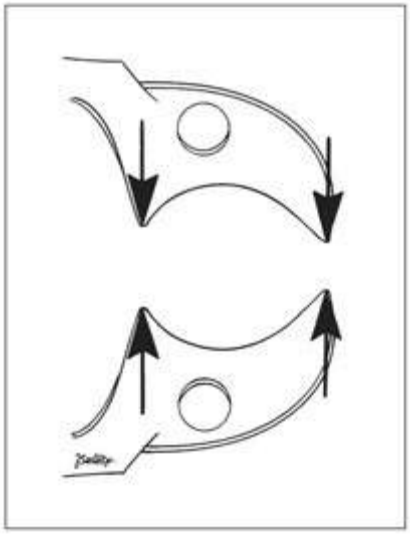


Fig 8-22 Rubber dam clamps should contact the tooth at the mesial and distal extent of the jaws. This four-point contact provides stability, or resistance to rotation or dislodgment, for the clamp.

The stiffness of the bow of the clamp should also be maintained. Clamps should be expanded with the clamp forceps no more than is necessary for the clamp to be passed over the facial and lingual heights of contour of the tooth. If a clamp has been overexpanded, it will grasp the tooth with less strength and is more likely to be dislodged. Occasionally, the jaws of clamps that have been overexpanded may be squeezed together so that enough of the strength returns, but it is usually best to discard a clamp that has been overexpanded.

Floss ligatures. Many clinicians and dental schools recommend that dental floss be attached to every clamp used in the mouth to allow retrieval if the clamp is dislodged or breaks. Certainly, it is wise to attach floss to the clamp that is

positioned in the mouth prior to application of the dam. After dam placement is completed, however, the floss causes leakage if it extends under the dam or is in the way if left to dangle in the operating field. A solution is to attach the floss to the clamp during application of the dam (see Fig 8-35a) and to cut and detach the floss from the clamp after the dam is in place. If the clamp dislodges or breaks after the dam is in place, it will either be catapulted from the mouth by the tension of the dam or be trapped by the dam so that it cannot be swallowed or aspirated.

When a winged clamp is attached to the dam during placement of the clamp onto a tooth, the attachment of a floss ligature to the clamp is redundant. Floss also need not be attached to a second clamp placed for retraction after the dam is in place.

Other retainers

Other methods are sometimes used for rubber dam retention:

- Dental floss or tape is placed doubly through a contact and then cut to a short length so that it does not impede access (Fig 8-23a).



Fig 8-23 Alternative methods for dam retention. *(a)* Dental tape placed doubly through the contact distal to the distalmost exposed tooth. *(b)* Short strip of rubber dam used to retain the dam. *(c)* Anesthetic cartridge plunger tied around the distal tooth with floss. *(d)* Elastic cord used as a rubber dam retainer (see Fig 8-34).

- A short strip of rubber dam material is cut from the edge of the rubber dam, stretched and carried through the contact, and then allowed to relax to retain the dam (Fig 8-23b).
- Floss is tied to a sterilized rubber plunger from an anesthetic cartridge or similar item and then tied around the most distal isolated tooth (Fig 8-23c).

- Elastic cord, eg, Wedjets (Hygenic), is placed interproximally to retain the dam (Figs 8-23d; see also Fig 8-34).

Modeling compound

Modeling compound may be used as an adjunct to the application of any clamp as a retainer or retractor. It is especially useful and necessary for anchoring and stabilizing the no. 212SA retainer (see Figs 8-21b and 8-21e).

For stabilizing a clamp, use of modeling compound, either red or green, is recommended. The clamp is positioned appropriately on the tooth and held in position with a finger until stabilization is completed. There are several effective methods for applying compound to stabilize a clamp. One of these techniques allows the practitioner to have prewarmed modeling compound immediately available and avoids the use of a flame. With this technique, modeling compound is placed into a plastic syringe, such as a large Monoject or impression syringe, which is then placed in a water bath at the appropriate temperature for the type of modeling compound used. The diameter of the aperture of the tip should be made larger to allow the softened compound to be ejected easily from the syringe. When the clamp is positioned, the practitioner removes the syringe with the prewarmed modeling compound from the water bath and flows it into the desired area to stabilize the clamp.

Another technique involves the use of the compound in stick form. A stick is held over a low alcohol flame and rotated and moved back and forth so that the length to be softened is heated evenly (Fig 8-24a). After the surface is softened, the stick is withdrawn from the flame to allow the heat to diffuse to the center of the stick. When the length is warmed to the center, there will no longer be a core of unsoftened compound to support the shape, and the softened length will sag or droop (Fig 8-24b). If the stick has been overheated, so that it elongates in addition to drooping, it should be cooled slightly in a container of water. Before the compound is taken to the mouth, the surface should be briefly reheated to enhance adhesion of the compound to the retracting clamp and teeth.

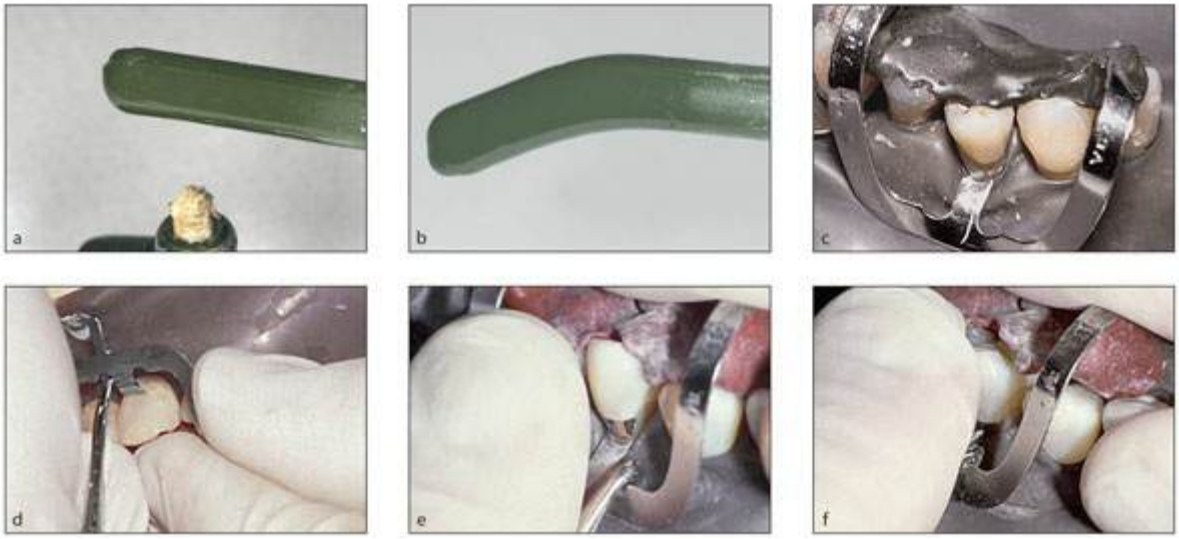


Fig 8-24 Use of modeling compound to stabilize a no. 212SA retractor. *(a)* The compound stick is warmed in an alcohol flame. *(b)* After warming, the stick is removed from the flame and held until the heat has diffused to the center of the stick so that the warmed end of the stick begins to droop. *(c)* Use of compound to stabilize a no. 212SA retractor is completed. *(d)* Removal of the clamp with forceps may be hampered by the compound on the lingual aspect. If so, a facial notch of the clamp may be engaged with an instrument such as a plastic instrument. *(e and f)* A no. 0 crochet hook (with handle made of laboratory acrylic) or a no. 34 surgical elevator modified to mimic a no. 0 crochet hook may be used to engage a notch of the clamp. Such a modified surgical elevator is manufactured by Hu-Friedy. The facial jaw of the clamp is then pulled facially away from the tooth surface and rotated occlusally or incisally to quickly remove the clamp and the stabilizing compound.

The compound should be applied to the retainer and teeth in a location as far away from the area to be restored as possible. The stick is then twisted and pulled away, leaving softened compound in place. The compound should be shaped and molded with damp, gloved fingers into embrasures and made to contact a large area of the clamp and the lingual surfaces of the teeth. It should then be cooled with the air syringe for 20 seconds or more. Stabilization of the retracting clamp is then completed (Fig 8-24c); the finger holding the clamp may now be released, and the clamp is tested for stability.

Compound should be kept away from the planned area of operation so that it will not inhibit access; in that regard, for a facial restoration, compound should be confined to the occlusal (or incisal) and lingual surfaces. Full advantage should be taken of the lingual surfaces for maximum dependability of attachment of the compound to the teeth. When the lingual surfaces are covered by the compound, the lingual notches for the clamp forceps will be covered. To remove the clamp with forceps, the operator would have to chip away the compound to expose a lingual notch. In a simpler method, an instrument is used to pull the facial jaw of the clamp

away from the facial surface and then occlusally (incisally) (Figs 8-24d to 8-24f).

Inverting instrument

Almost any instrument may be used for inverting the dam. Commonly used instruments include explorers such as the no. 23 (Fig 8-25a), plastic filling instruments such as the no. 1-2 (Fig 8-25b), or a beavertail burnisher (Fig 8-25c). Dental tape or floss used interproximally is also useful for dam inversion. Inversion of the dam is described later in this chapter and illustrated in Fig 8-38.



Fig 8-25 Inverting instruments. Note the tip of the air syringe in each instance; a high-volume stream of air is used to dry the tooth and dam surfaces to facilitate inversion. *(a)* No. 23 explorer. *(b)* No. 1-2 plastic instrument. *(c)* Beavertail burnisher.

Wedge

The wooden wedge, which is used to stabilize a matrix and hold it against the gingival margin of a cavity preparation involving a proximal tooth surface, is also useful for protecting the dam (Fig 8-26) when rotary cutting instruments are used in proximal areas. Placement of water-soluble rubber dam lubricant on the wedge enhances the ease of wedge placement.



Fig 8-26 Wooden wedges are used to protect the dam from being cut during a procedure that involves the use of burs or cutting instruments near the dam.

Scissors

Scissors are often useful in preparing the dam for insertion and are a necessity for cutting the dam for removal. Blunt-ended scissors are preferred by many operators,

but other scissors, such as sharp crown and collar scissors and Quimby scissors (see Fig 8-41b), will also serve well. Scissors used for cutting rubber dams must be sharp, or they will frustrate the operator.

Dental tape and floss

Waxed tape or floss, not unwaxed floss, is recommended for flossing the dam through interproximal contacts. Waxed tape, or ribbon floss (see Fig 8-35f), will carry more of a septum through a contact in a single pass than will the narrower floss, but the tape must be maintained flat and not bunched up, or it will be difficult to pass through the contact.

Interproximal contact disk

An interproximal contact disk (Thierman Products or Centrix) is used to plane rough enamel, amalgam, or resin composite contacts so that the floss will pass through without shredding and so that the dam can be flossed through without tearing (Fig 8-27). The plain metal disk, without abrasive, is recommended. This instrument should not be used in a contact that involves a gold casting because it can cut into the gold and produce additional obstruction to passage of the floss through the contact.

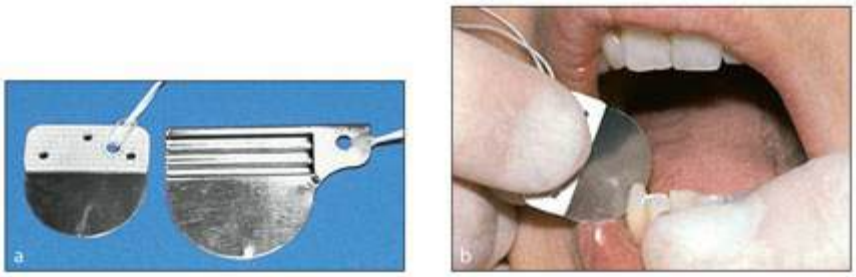


Fig 8-27*(a)* Interproximal contact disks or planes. The disk with the white plastic handle is from Thierman Products, and the disk with the metal holding area is from Centrix. *(b)* An interproximal contact disk is used to plane a rough contact.

The disk is placed into the occlusal embrasure and rocked facially and lingually as it is pushed firmly, but with control, gingivally. If it cannot be worked through the contact, the teeth should be separated slightly with a plastic instrument placed snugly into the gingival embrasure and rotated slightly while the disk is being pushed into the contact from the occlusal embrasure. Several passes of the disk through the contact will usually plane it smooth.

Lubricant

Rubber dam lubricant makes a significant difference in the ease with which the dam is applied. A water-soluble lubricant is preferred. A product that has proven especially suitable for lubricating the rubber dam is Velvachol water-miscible vehicle (Healthpoint) (Fig 8-28). Velvachol is a pharmaceutical product manufactured as a water-soluble ointment base, but it is an excellent dam lubricant. Petroleum-based lubricants, such as Vaseline (Chesebrough-Pond's), should be avoided as rubber dam lubricants because they are difficult to remove from the dam after application and therefore can impede bonding procedures and make inversion of the dam more difficult.



Fig 8-28 A water-soluble rubber dam lubricant, such as Velvachol, can be loaded into a syringe, such as a 3-mL disposable syringe. The lubricant can be dispensed from the syringe onto the tissue surface (underside) of the rubber dam or onto a glove for coating of the dam adjacent to the holes.

Water-soluble lubricant is applied in a thin coat in the area of the holes on the tissue surface of the dam before it is taken to the mouth (Fig 8-29). The lubricant makes passage of the dam through the interproximal contacts much easier, and the dam will often pass through the contacts in a single layer without the use of floss. If additional lubrication is desired, lubricant may be applied to the teeth prior to placement of the dam.

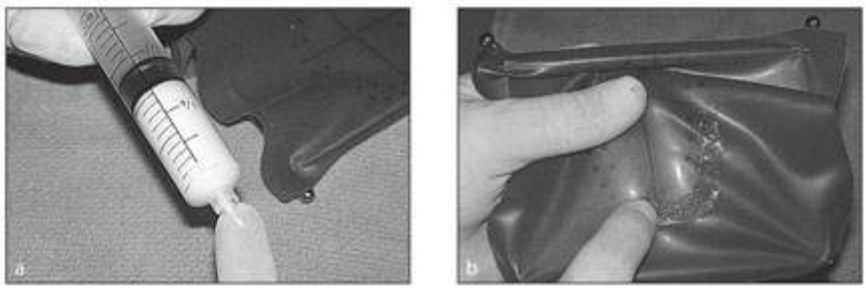


Fig 8-29*(a)* Water-soluble lubricant can be carried to the dam with a finger. *(b)* The dam lubricant is layered on the tissue surface of the dam in the area of the holes.

A lubricant for the lips will make the patient more comfortable during the procedure. Petroleum-based lubricants, such as Vaseline, cocoa butter, silicate lubricant, or lip balm, function well as lip lubricants.

Application and removal

Preparation of the mouth

Teeth should be cleaned, if necessary, and contacts should be checked with floss. The rapid passage of dental floss through each contact that will be involved in the isolation is very important and, if accomplished as a part of the routine, will save chair time. Any rough contact should be smoothed with the interproximal contact disk (see Fig 8-27b), not only to facilitate dam placement but also to enable the patient to clean each interproximal area during routine flossing.

If a restorative procedure that involves an occlusal surface is planned, maximum intercuspation contacts may be marked with articulating paper or tape prior to application of the dam. Centric occlusion markings may be coated with a clear light-cured resin or varnish to protect them from being rubbed off. An applicator or brush containing the liquid resin should be touched to the enamel adjacent to the markings and the material allowed to flow across the markings prior to curing (Fig 8-30).



Fig 8-30 The centric occlusion markings were protected by a varnish or light-cured resin during dam placement; had the markings not been protected, the placement procedure would likely have erased them.

If the lips are to be lubricated, this should be accomplished prior to application of the dam (Fig 8-31).

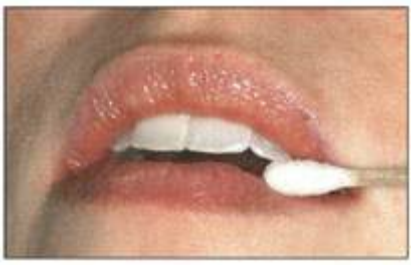


Fig 8-31 Lips are lubricated with petroleum-based lubricant prior to placement of the rubber dam.

Preparation of the dam

Use of a prestamped, dark (gray, green, or blue), heavy (or extra heavy) gauge dam

material is recommended. Various hole sizes should be used to ensure a seal around the variety of tooth sizes (Fig 8-32). For example, an Ivory punch has six hole sizes, numbered 1 (smallest) through 6 (largest) (see Fig 8-4). For standard latex heavy-gauge dams, recommended hole sizes are 5 for clamped molars; 4 for other molars; 3 for premolars, canines, and maxillary central incisors; and 2 for maxillary lateral incisors and mandibular incisors.

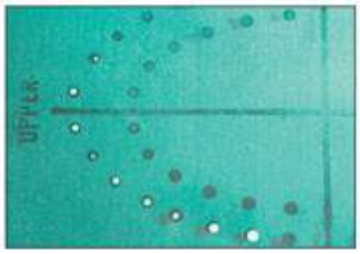


Fig 8-32 Varying hole sizes are used to seal the dam around various sizes of teeth.

Some variation from the recommended hole sizes may be needed, depending on the size of individual teeth, operator preference, dam material, and gauge of the dam, but a range of hole sizes should be used to prevent leakage between the dam and the teeth.

For operative procedures involving posterior teeth, the tooth or teeth to be restored should be exposed, as well as at least one tooth posterior to the most distal tooth to be restored, if possible. In addition, all teeth around to the central or lateral incisor on the opposite side of the same arch should be exposed. This extension of the area of isolation to the opposite side will hold the dam flat in the arch to give room for fingers and instruments in the area of the teeth to be restored. It will also expose teeth in the anterior area for finger rests during the operation (Fig 8-33).



Fig 8-33 The incisal edges of the anterior teeth are used as a finger rest.

For anterior restorations, exposure of the first premolar to the first premolar on the opposite side of the arch is recommended (Fig 8-34). This will provide room for the mirror and for hand instruments on the lingual aspect of the anterior teeth.



Fig 8-34 For an anterior restoration, the first premolars are exposed to provide anchorage for the dam and to leave adequate working room on the lingual aspect of the anterior teeth.

When a prestamped dam or a template is used, holes should be punched away from the spots to accommodate atypical alignment of teeth. In addition, when the dam is being prepared to provide isolation for Class 5 restorations, the hole for the tooth to receive a facial Class 5 restoration should be punched approximately 1 mm facial to the spot (see Fig 8-21c) to allow retraction with the no. 212SA clamp. No holes should be punched for missing teeth.

After the dam is punched, the tissue side of the dam should be lubricated with a water-soluble lubricant. A small dollop of lubricant is applied to the tissue surface and smeared over the surface of the dam in the area of the holes (see Fig 8-29b). The rubber dam frame can then be attached to the top and bottom of the dam, leaving a relaxed area or "pouch" of dam material between the top and bottom (see Fig 8-35b). Attaching the dam to the frame in this way holds the edges of the dam away from the holes for better visualization during application.

Placement of the dam

If a local anesthetic agent has been administered to provide pulpal anesthesia for the tooth or teeth being restored, at least a portion of the gingival tissue will have also been anesthetized. If an inferior alveolar block has been given, the lingual nerve will almost always have been anesthetized as well, so the gingival tissue lingual to the mandibular posterior teeth will also have been anesthetized. If infiltration anesthesia has been administered to maxillary teeth, the facial gingival tissue will have been anesthetized. For application of a rubber dam clamp, the portions of the gingival tissue that have not been anesthetized along with the delivery of pulpal anesthesia will not normally need to be anesthetized. When the clamp is applied, as long as the points of the clamp's jaws are firmly on the tooth and have not penetrated gingival tissue, the patient may feel some discomfort for a few seconds where the jaws are pressing against tissue. This pressure discomfort will usually disappear within 1 minute due to "pressure anesthesia," and injection anesthesia for the gingival tissue

is usually unnecessary. If additional gingival anesthesia is necessary, topical anesthetic solutions or gels may suffice.

When the clamp is applied to the tooth with the clamp forceps, the clamp should be expanded only enough to allow it to pass over the crown of the tooth. Overexpansion of the clamp will permanently distort it so that it will be weak, unstable, and more likely to dislodge from the tooth.

Dam over clamp. A wingless clamp is placed on the tooth. It is recommended that a finger be maintained over the inserted clamp to prevent its dislodgment until its stability on the tooth has been confirmed. The operator checks stability by engaging the bow of the clamp with an instrument and firmly attempting to pull it occlusally (Fig 8-35a). If the clamp rotates on the tooth, it is not stable and should be repositioned or replaced.

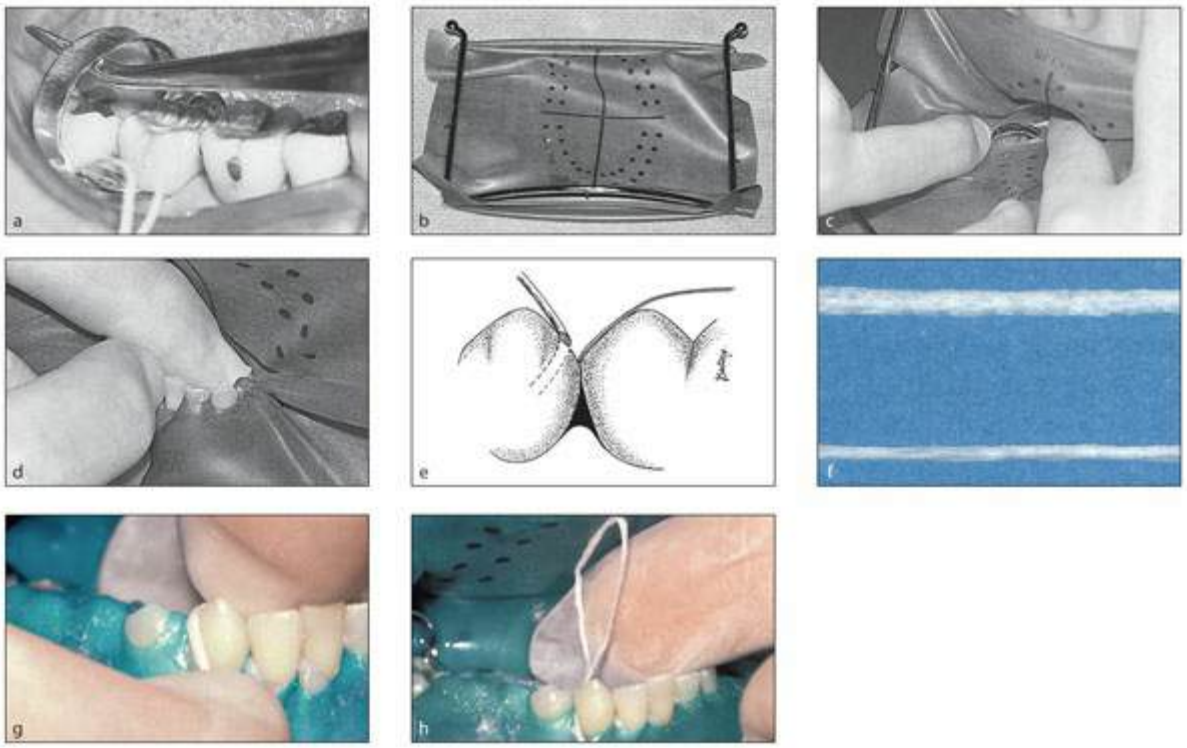


Fig 8-35 Dam over clamp method of dam application. *(a)* The clamp (modified no. W8A) is tested for stability. To do so, the operator attempts to pull the bow occlus-ally. *(b)* The dam is fitted loosely on the frame. *(c)* The distal hole of the dam is carried over the bow of the clamp. *(d)* The septa are worked through anterior contacts as a gloved fingernail is used to slightly separate teeth. *(e)* The leading edge of the dam is touching the occlusal aspect of the interproximal contact; floss is on the adjacent tooth. *(f)* Waxed dental tape, or ribbon floss *(top)*, if it is not folded or bunched, will carry more of the dam septum through the contact in a single pass than will waxed floss *(bottom)*, but either type will serve the purpose. *(g)* The dam septum is lying on the mesial aspect of the mandibular first premolar, with its leading edge at the mesial contact; the floss is lying on the distal aspect of the canine, ready to move to the contact, meeting the dam there. The floss will then carry at least a portion of the septum through the contact. *(h)* The floss has been doubled

back to the facial aspect and passed through the contact again, carrying another portion of the septum through the contact. The floss is then removed from the contact; one or both of the tails of the floss are pulled facially away from the teeth.

The top and bottom attachment points of the Young frame are engaged at the top and bottom of the dam to give a slackness or pouching of the dam (Fig 8-35b). The tissue side of the dam is lubricated in the area of the holes. Then, with a finger on each side of the distal hole in the dam, the dentist (or assistant) stretches the dam so that the hole is enlarged and appears to be an open slit; the hole is then carried over the bow and jaws of the clamp (Fig 8-35c). The hole at the opposite end of the row (usually for the lateral or central incisor on the opposite side) is then passed over the appropriate tooth, and the septa are worked through the interproximal contacts.

A gloved fingernail used to slightly separate the anterior teeth is very helpful, and floss is not usually needed to carry the dam through anterior interproximal contacts (Fig 8-35d). To use the "fingernail technique," the edge of the septum is positioned at the incisal extent of the contact and pulled gingivally with fingers on the facial and lingual aspects. This method also can frequently facilitate septum passage through interproximal contacts of posterior teeth.

Good lubrication of the dam is necessary for easy and quick application. The dam should be passed through each contact in a single layer. This may be accomplished by stretching a septum over one of the teeth adjacent to the contact and sliding the edge of the rubber to the contact so that a leading edge of dam is touching the contact (Fig 8-35e).

In posterior areas, the leading edge should be touching the occlusal portion of the contact in the occlusal embrasure. Waxed tape (ribbon floss) or waxed floss may then be used to move the dam progressively through the contact (Figs 8-35e to 8-35g). Tape will carry more of the rubber through the contact in a single pass than will floss. If tape is used, like the rubber, it should be taken through the contact in a single layer, not twisted or bunched up.

If the dam goes through with one pass of the floss, the floss should be removed from the contact without pulling the rubber back out. To accomplish this, the tail of the floss that is on the lingual side of the teeth is doubled back across the occlusal embrasure of the contact so that both ends are on the facial aspect; then the tape is pulled facially through the contact. If only a portion of the septum goes through the contact with the first pass of the floss or tape, the floss should be doubled back and passed through the contact again; it is then pulled facially out of the gingival embrasure (Fig 8-35h). The tape should be passed through repeatedly until the entire

septum has been carried through the contact.

Winged clamp in dam. Prior to lubrication of the dam, the clamp is placed into the distal hole so that the hole is stretched over the wings of the clamp from its tissue side (Fig 8-36a; see also Fig 8-16). The dam is then lubricated, and the frame is attached. The forceps are inserted into the holes of the clamp, and the clamp, dam, and frame are carried as a unit into place (Figs 8-36b and 8-36c). After the stability of the clamp is confirmed, the dam material on the wings of the clamp is pulled off the wings with finger tension or with a bladed instrument such as a plastic instrument. The remainder of the dam is placed as previously described (Fig 8-36d).



Fig 8-36 Winged clamp in dam method of dam application. *(a)* A winged clamp (no. 27) is inserted into the distal hole of the dam. *(b)* The clamp–dam frame assembly is carried to the mouth as a unit. *(c)* The clamp is placed on the mandibular second molar. *(d)* The dam has been applied to the quadrant, and a no. 1-2 plastic instrument is used to pull the edges of rubber off the wings of the clamp.

Wingless clamp in dam. The distal hole of the lubricated dam is passed over the bow of a wingless clamp, such as the modified no. W8A, so that the hole comes to rest at the junction of the bow and the jaw arms (Fig 8-37a). The frame is not attached to the dam at this point. The dam is gathered up and elevated to expose the jaw arms of the clamp, and the forceps are then inserted into the forceps holes (Fig 8-37). The gathered dam is carried to the mouth with one hand and the forceps with the other. After the clamp is applied to the distal tooth and the dam has been pulled over the jaws of the clamp, the frame is attached and the other teeth are isolated as previously described.

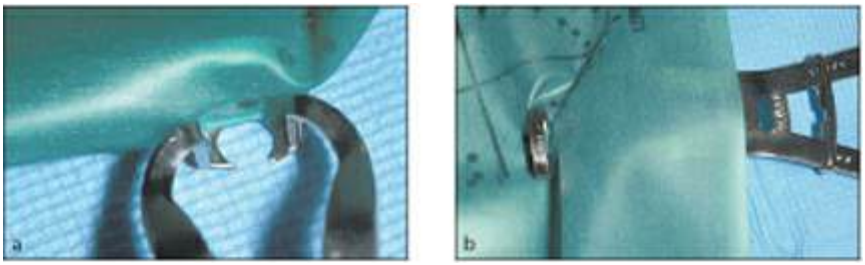


Fig 8-37 Wingless clamp in dam method of dam application. *(a)* The clamp in the dam. *(b)* Dam and clamp with forceps in place.

Clamp after dam. The dam is applied to the teeth and then the clamp is placed. This technique, occasionally necessary, is the most difficult.

Completion of application

Application of the napkin. For longer procedures, the use of a rubber dam napkin is recommended. The napkin may be positioned before or after the dam is in place on the teeth. For placement of the napkin after the dam has been applied, the frame is removed, the napkin is placed so that its edges remain on the skin and not in the mouth, and the frame is replaced.

Adjustment of the dam in the frame. The frame and dam are adjusted so that there is a minimum of folds and wrinkles and so that the dam does not obstruct the nostrils.

Washing of the dam. The dam and isolated teeth are washed with an air-water spray to remove the lubricant. After they are washed, the dam and teeth should be dried with air from the air syringe.

Inversion of the dam. The dam should be inverted around the necks of the teeth, at least in the area of the tooth or teeth to be restored. The edge of the dam that is against the tooth acts as a valve. If the edge is directed occlusally (Fig 8-38a), when a positive pressure is created by the tongue and cheeks under the dam, the valve opens, and saliva and other liquids under the dam are pushed between the tooth and dam to flood the operating field; then, when a negative pressure is created under the dam, the valve closes and the saliva is trapped in the field. When the dam is inverted, a positive pressure under the dam simply serves to push the valve more tightly against the tooth (Fig 8-38b) so that no flooding of the field occurs.

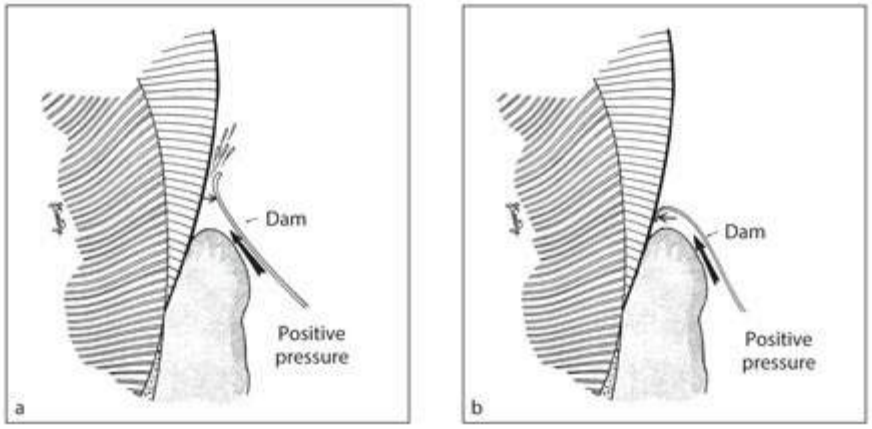


Fig 8-38*(a)* Without inversion of the dam, positive pressure under the dam, created by tongue movement, swallowing, etc, will cause leakage of saliva into the operating field. *(b)* With inversion of the dam, positive pressure under the dam only causes the dam to seal more tightly against the tooth, preventing leakage.

Almost any instrument may be used to tuck the edge of the dam gingivally (see Fig 8-25). A steady, high-volume stream of air should be directed at the tip of the instrument used to invert the dam, and the instrument should be moved along the margin of the dam so that the inversion is progressive.

Floss may be used to invert the dam in interproximal areas (Fig 8-39a). When it is used to carry the edge of the dam gingivally, the floss should not then be pulled occlusally for removal because it will frequently pull the edge of the dam with it, eliminating the inversion. Instead, the floss can be doubled over on itself on the lingual aspect and passed again through the contact. Then one end is pulled in a facial direction so that the floss rolls from the sulcus, leaving the dam inverted (Fig 8-39b). In this floss-facilitated inversion, a steady stream of air is as helpful as it is when inversion is accomplished with an instrument. The dam inverts more easily when the surfaces of the tooth and adjacent dam are dry.

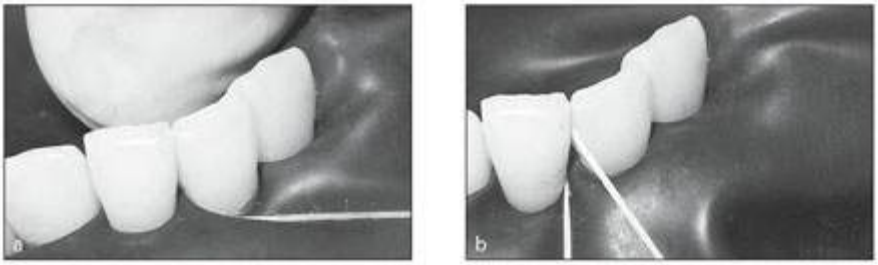


Fig 8-39*(a)* Floss is used to invert the dam in an interproximal area. *(b)* Floss is "rolled out" to the facial aspect to prevent reversal of the inversion it accomplished.

Ligation for retraction

Occasionally, because of the gingival location of a lesion or preparation, a ligature

may be helpful for retracting the dam to a position cervical to the margin. Dental floss or tape is passed through one interproximal area, around the lingual aspect of the tooth, then back through the other interproximal area or contact. Then the floss or tape is tied, preferably with a surgeon's knot, on the facial aspect of the tooth. This ligation will usually carry the edge of the dam cervically to expose the area of the planned margin (Fig 8-40). After the ligature is tied securely, the tails of the floss or tape may be pulled to some portion of the rubber dam frame and attached to it, or they may be shortened to prevent their getting in the way of the operation. Prior to removal of the dam, the ligature should be cut with scissors or a sharp carver or scalpel blade and removed.

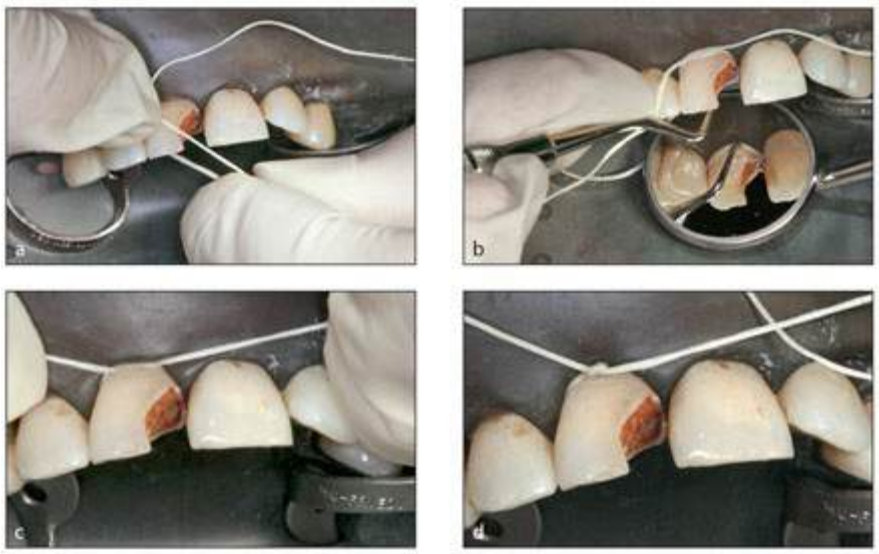


Fig 8-40 Dental floss is used to ligate the rubber dam for retraction of the dam cervical to a planned preparation. *(a)* The floss is first passed around the tooth. *(b)* It is then pushed cervically past the cingulum on the lingual aspect. *(c and d)* It is tied to secure it in a cervical position. (Courtesy of Joseph Connor, San Antonio, Texas.)

Protection of the dam

Torn dams provide poor isolation, so expenditure of a little effort to prevent tearing is worthwhile. An example of protection would be the use of a wedge interproximally when rotary instruments are used in the proximity of the dam (see Fig 8-26). Another example is the use of a second clamp to retract the dam below a margin that is near, or below, the level of the gingival crest (see Fig 8-47).

Removal of the dam

The interproximal septa are stretched and clipped with scissors (Fig 8-41). The scissors are held so that the tips are not in contact with any tissue (see Fig 8-41b).

When all septa are cut, the clamp is removed with the forceps and the dam is snapped from the teeth.



Fig 8-41*(a)* To remove the dam, the interproximal septa are stretched for cutting. *(b)* One blade of the scissors is used to pull the dam well away from any tissue before the septum is cut.

After the dam is free from the mouth, the teeth should be examined to ensure that no rubber remains around them or in the contacts. The frame should be removed from the dam, and the dam should be laid flat on a surface and examined to ensure that no pieces are missing (Figs 8-42a and 8-42b). If a piece is missing and unaccounted for, the mouth should be reexamined in the area of the missing piece of dam; any remnant should be removed (Fig 8-42c). A small piece of dam left subgingivally can cause inflammation, gingival abscess, or even significant loss of periodontal support.



Fig 8-42*(a)* While the dam is on the frame, it is difficult to determine if any portion is missing. *(b)* The dam is removed from the frame and laid on a flat surface. Note that a portion of dam is missing. *(c)* The missing piece is located in the mouth and removed.

Special considerations

Bite block

Patients often have difficulty keeping their mouths open or are uncomfortable with wide opening. A rubber bite block can relieve their discomfort, allow them to relax the musculature, and permit them to keep the mouth open without effort. Bite blocks are available in a variety of sizes (Figs 8-43a and 8-43b). A piece of floss or tape may be attached to the bite block to allow retrieval if necessary (Fig 8-43c). Figure 8-44 shows placement of the bite block after the dam is in place.



Fig 8-43*(a and b)* Rubber bite blocks are available in various sizes. *(c)* Floss is attached to the bite block for emergency retrieval if necessary.

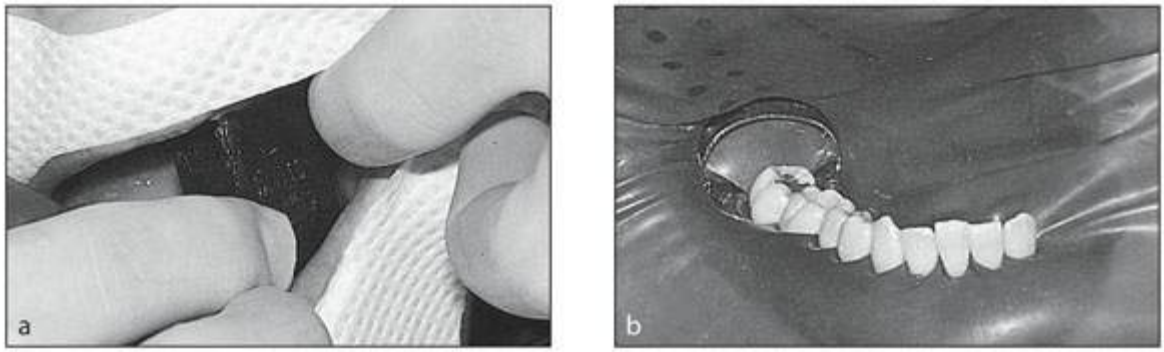


Fig 8-44*(a)* A bite block is inserted on the patient's left side after a rubber dam is applied to isolate the right quadrant. *(b)* The dam with a bite block in place aids access to the field and increases patient comfort.

Isolation for a fixed partial denture

Whenever possible, it is best to achieve isolation without incorporating a fixed partial denture into the isolated operating field. When a fixed partial denture must be included, there are several techniques that can be used; they are all somewhat time-consuming but often valuable. Two methods are described.

Cyanoacrylate method. Figure 8-45 illustrates this method. Holes for the teeth are punched in the dam. The holes for the abutment teeth are connected with a cut that is in an arc to give a "tongue" of dam material between the holes. The tongue of material is folded back, and a piece of dam material is attached with cyanoacrylate glue over the opening left when the tongue was folded back. This piece is glued into place so that there is a slit connecting the abutment holes and a tongue of material that is free to swing down over the attached piece of dam material.



Fig 8-45 Rubber dam isolation around a fixed partial denture (cyanoacrylate method). *(a)* A four-unit fixed partial denture extends from the mandibular first premolar to the second molar. *(b)* The holes for the abutment teeth are connected with an arced cut. Note the small piece of dam material at bottom that will be used as shown in Figs 8-45c and 8-45d. *(c and d)* A small piece of dam is glued in place. *(e)* A no. W8A clamp is positioned on the second molar. *(f)* The dam is carried to place. *(g)* The tongue of dam material is tucked under the pontics with a periodontal probe. *(h)* The tongue of material is grasped and pulled lingually with a hemostat. *(i)* Glue is applied for attachment of the rubber dam tongue. *(j)* The tongue of dam material is held in place with a hemostat and a cotton-tipped applicator while the glue sets. *(k)* Isolation is complete.

The dam is inserted over all teeth for which holes have been punched, and the tongue of material is pulled under the pontic(s) and glued into place on the added piece of dam. Tension on the tongue while the glue is setting (10 to 15 seconds) will ensure that the dam is tight around the abutments after tension is released.

Ligation of septa around the retainer-pontic connectors. This method is illustrated in Fig 8-46. This procedure is for three-unit fixed partial dentures or splinted teeth. Holes are punched for each abutment, and, for three-unit fixed partial dentures, another hole is punched for the pontic. A piece of floss or suture material

is used to tie through the holes so that the septum between adjacent holes is stretched around the retainer-pontic connector. If floss is used, a "floss-feeder," made for carrying floss under partial denture pontics for oral hygiene measures, may be used to guide the floss under the pontic and pull it through. If suture material is used, the suture needle may be blunted and used for that purpose.

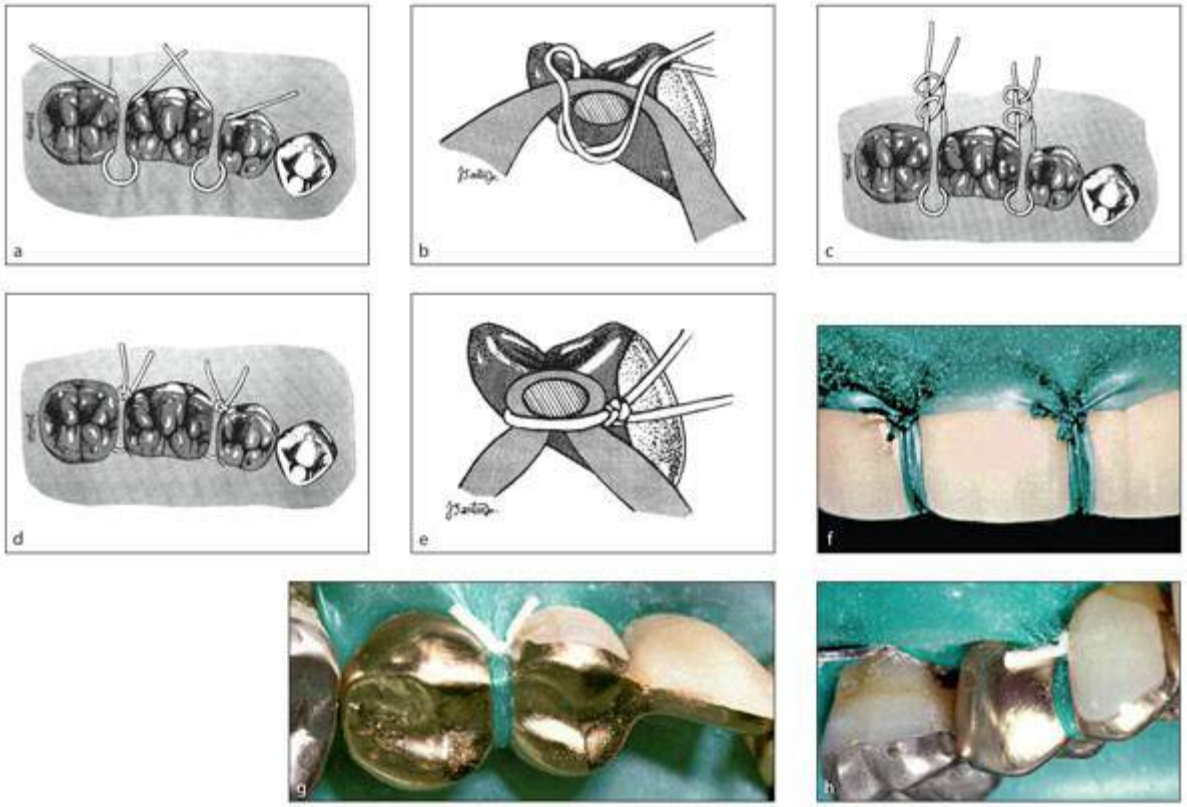


Fig 8-46 Isolation around a three-unit fixed partial denture or splinted teeth (ligation method). *(a)* Holes are punched for the abutment teeth and pontic, and the dam is positioned. The septa on the mesial aspect of the mesial abutment and the distal aspect of the distal abutment are flossed to place, and then the holes are stretched over the abutments and the pontic. *(b)* A ligature is threaded through an abutment hole on the facial aspect, under the retainer-pontic connector, through the same hole again on the lingual aspect, around the septum, through the pontic hole on the lingual aspect, back under the connector to the facial aspect, and back through the pontic hole. *(c to e)* The ends of the ligatures are tied together to pull the rubber septum tightly around the connector. *(f)* Sutures have been used for field isolation involving an anterior three-unit fixed partial denture. *(g and h)* Floss has been used for isolation involving a cantilevered canine pontic attached to splinted premolars.

Use of multiple clamps

In addition to the clamp on the distal tooth, which retains the posterior portion of the dam, a second (or third) clamp is often needed. When the no. 212SA or other butterfly clamp (retractor) is used to retract tissue and dam for a Class 5 or other restoration, it is almost invariably used in addition to the posterior clamp. If a cavity

that is at least partly subgingival is to be prepared, a clamp on that tooth will prevent the dam from riding up over the margin (Fig 8-47).



Fig 8-47 A clamp is used to retract the dam and give access to the gingival extent of a root restoration on this maxillary central incisor. A no. W1A clamp was used because of the apical inclination of its jaws.

Placement of clamp over dam

When it is desirable to clamp a tooth that was not considered when the dam was punched, the clamp may be applied over the dam (see Fig 8-15). The clamp jaws should be dull, so as not to cut through the dam, and the dam should be stretched loosely over the tooth being clamped, as stretching it tightly will cause the clamp jaw to perforate the rubber, initiating a tear in the dam.

Gingival relaxation incisions

When using a no. 212SA retractor for isolation for a Class 5 restoration, the jaw of the retractor should be positioned at least 0.5 mm (preferably 1.0 mm) gingival to the gingival margin of the planned restoration. This can usually be accomplished without laceration of tissue, because the free gingiva is elastic enough to be retracted. If, however, the free gingival margin is fibrous and difficult to displace gingivally, forced retraction could lacerate the tissue. In such a case, it is preferable to make one or two small incisions[20,21] to allow the tissue to be displaced without tearing.

For this technique (sometimes referred to as a *miniflap procedure*) to be successful, the periodontium must be healthy. The incisions should be confined to the keratinized gingival tissue and kept as short as possible (just long enough to allow adequate exposure for isolation). Incisions can often be limited to the free gingiva, and, although reattachment to previously unexposed cementum can be expected, unnecessary severing of attachment should be avoided. Full-thickness vertical incisions should be initiated at the mesial and/or distal aspects of the facial surface

and should be directed perpendicular to the root and surface of bone, first perpendicular to the gingival margin and slightly toward the interproximal papilla, then apically (Fig 8-48).

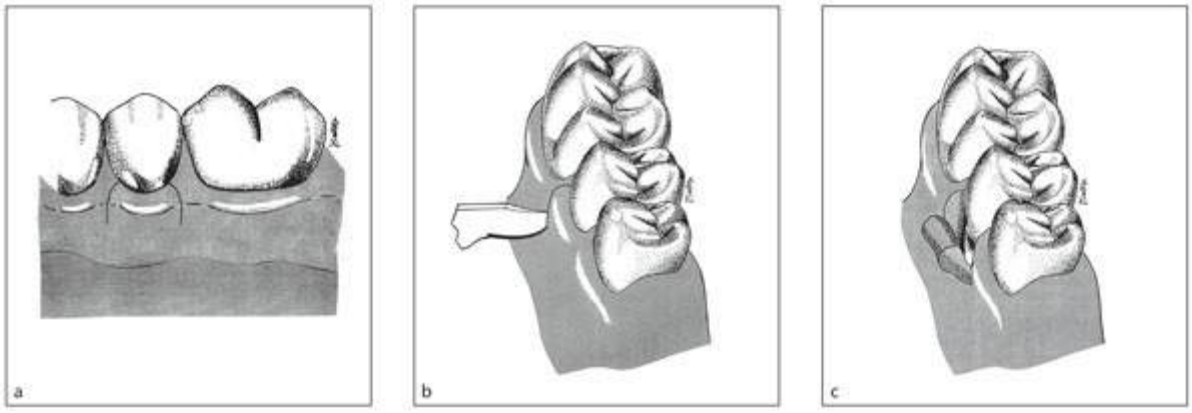


Fig 8-48*(a)* Gingival relaxation incisions are made within the keratinized gingival tissue. Either one or both can be made, depending on the amount of release needed for relaxation of the tissue. *(b)* The no. 15 scalpel blade is used to make the incision. *(c)* The tissue flap is reflected away from the root prior to application of the dam and no. 212SA retractor. The incisions are directed slightly into the papilla and then vertically.

The blade of a plastic instrument or a beavertail burnisher may be used to push the tissue and rubber dam back while the facial jaw of the no. 212SA clamp is being situated on the root of the tooth. Again, the jaw should be dull, not sharp, so that it will not damage the root surface. A finger should be used to hold the clamp in place while it is stabilized with compound (see Figs 8-24a to 8-24c). After the restorative procedure is completed, the no. 212SA clamp is removed (see Figs 8-24d to 8-24f), then the dam is removed. Any blood in the area is washed away. The reflected gingival tissue is returned to its original location and held there with a dampened gauze sponge and finger pressure for about 2 minutes to allow initiation of a fibrin clot. As long as the incisions were confined to keratinized tissue, no sutures or periodontal dressing are needed, and healing should proceed uneventfully. See chapter 15 for more information on soft tissue management in conjunction with Class 5 restorations.

Evacuation of fluid from the dam

If the dentist must work without an assistant, a very effective method for evacuation of fluid from the rubber dam involves the use of a suction tube anchored within the operating field. One evacuation method that uses a readily available item and is quick and easy involves the modification of a saliva ejector, as described by Lambert.[22] The molded plastic tip is cut off with a pair of crown scissors; then an

additional 0.4 inch of the plastic tube is cut off without cutting the wire within the plastic. The 0.4-inch length of plastic tubing is then pulled off the wire, leaving the wire extending from the end of the tube (Fig 8-49a); using forceps such as hemostats, the wire is bent in its center at a 90-degree angle in the direction of the tube to form an "L" shape. The wire is then carried under the jaw of the clamp and placed into the hole in the jaw of the clamp, usually on the lingual side of the clamped tooth. The taut rubber under the clamp jaw will hold the wire in place in the hole and push the tube against the dam (Figs 8-49b and 8-49c). This method will supply continuous fluid evacuation during the operative procedure.



Fig 8-49*(a)* For unassisted fluid evacuation from the dam,[22] a saliva ejector is modified. First, the tip is cut off; then another 0.4 inch, except for the wire, is cut off, and the protruding wire is bent at a right angle (in direction of tube) to form an L-shaped hook. *(b and c)* For fluid evacuation, the wire hook is carried under the jaw of the clamp and allowed to go into the forceps hole of the rubber dam clamp.

Sealing a root concavity

The rubber dam seals well on convex tooth surfaces. If the dam is retracted so that its edge goes across a root concavity, however, saliva will leak into the operating field. A solution is to seal the gap between the edge of the dam and the concave root surface. This may be accomplished with a provisional restorative material, such as Cavit (3M ESPE), which hardens with moisture (Fig 8-50).

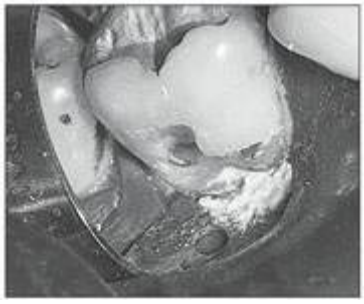


Fig 8-50 Sealing of the root concavity. The dam is retracted by the clamp to allow isolation for a large Class 5 restoration; the retraction is apical to the beginning of the root concavity of the furcation. The gap between the dam and the concave root surface is sealed with Cavit, a provisional restorative material that hardens when it comes into contact with moisture.

Repair of a torn rubber dam

A small tear in a dam may often be patched. A piece of dam material is cut to cover the tear, extending 1 cm or so beyond the tear on all sides. The piece is attached over the tear with cyanoacrylate glue.

Placement of a second dam over the first

If a dam is torn beyond repair during a procedure, the dentist might choose to remove the dam and replace it. Alternatively, another dam may be placed over the top of the first. Brownbill[23] recommended that this technique be used when there is leakage around teeth through incorrectly sized holes and when strong chemicals are to be used.

Latex allergies

There is an increasing awareness of latex sensitivity.[24–27] One survey[24] reported 3.7% of patients to have a latex allergy; the investigators recommended careful questioning of patients regarding a history of sensitivity to latex-based products so that the use of latex products, such as gloves and rubber dam, may be avoided with these patients.

For latex-sensitive patients, use of a latex dam should be avoided, as should other latex products. Nonlatex dam material is available from several manufacturers and should be on hand for latex-allergic patients. Current nonlatex dams have elastic properties very similar to latex. Some dentists have elected to use nonlatex dams exclusively for all their patients.

Summary of recommendations

Following are some of the protocols that facilitate rubber dam use:

- Use a heavy-gauge, prestamped dam.
- Floss through contacts prior to dam placement, planing any contact that shreds or tears the floss.
- Use a good water-soluble lubricant, such as Velvachol.
- Use a clamp designed for four-point contact on the tooth, and avoid overexpansion of the clamp so that the clamp will maintain its strength and will be stable as a retainer.
- Isolate enough teeth to hold the dam on the lingual aspect of the teeth away from the operating field and to provide exposed teeth for finger rests.

- With waxed floss, floss the dam through each interproximal contact in a single layer and avoid doubling or bunching the dam in the contact.
- Master the use of modeling compound to stabilize rubber dam retainers when necessary.

Other Methods of Isolation

Isolite

The Isolite (Isolite Systems) is a newer isolation device that provides illumination in addition to suction, retraction of tongue and cheek, and an integrated bite block (Fig 8-51). It is relatively simple to use and comfortable for the patient. Single-use mouthpieces come in six sizes (small, medium, and large regular; small, medium, and large DV [deep vestibule]). Mouthpieces conduct light so that it surrounds the teeth being isolated, and illumination is excellent. Many practitioners have incorporated the device in their practices. Like other devices that provide tongue retraction and suction, isolation is not as complete as that provided by the rubber dam. However, isolation with this device is excellent for many procedures.

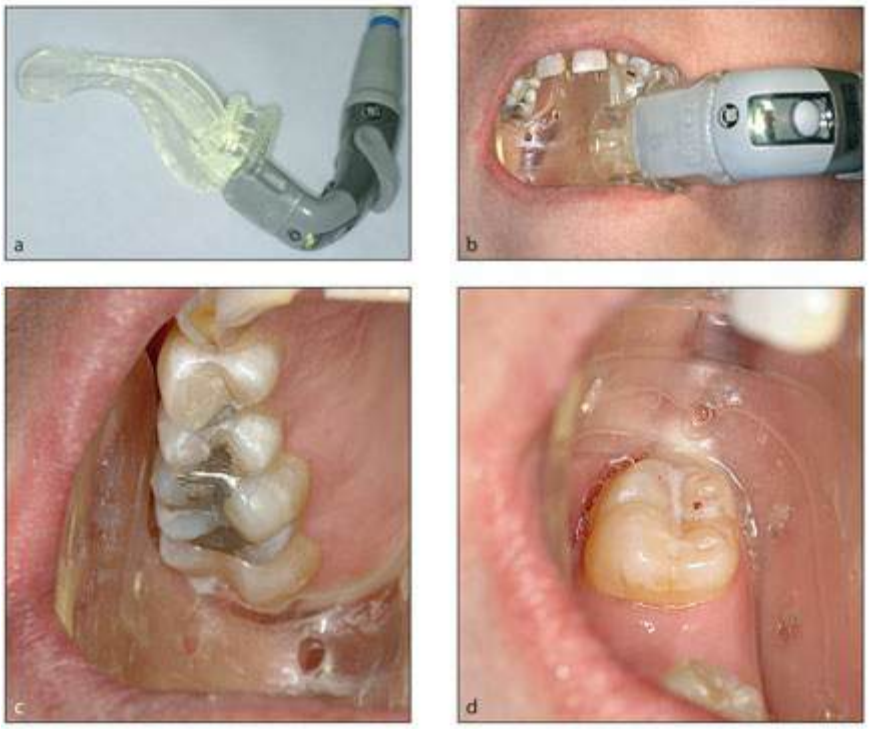


Fig 8-51 The Isolite provides tongue and cheek retraction, suction, an integrated bite block, and excellent illumination of the operating field. *(a)* Isolite device, with clear, flexible plastic, illuminated mouthpiece; *(b)* Isolite in place to isolate the right side of the mouth; *(c)* the isolated maxillary right posterior area; *(d)* the isolated mandibular right posterior area.

Svedopter

The Svedopter (Miltex) is a metal tongue-retraction device for isolation in mandibular posterior areas (Fig 8-52). It is designed so that the vacuum evacuator tube passes anterior to the chin and mandibular anterior teeth, over the incisal edges of the mandibular anterior teeth, and down to the floor of the mouth, to either the left or the right of the tongue. A mirrorlike vertical blade is attached to the evacuator tube so that it holds the tongue away from the field of operation. Several sizes of vertical blades are supplied by the manufacturer. An adjustable horizontal chin blade is attached to the evacuation tube so that it will clamp under the chin to hold the apparatus in place.

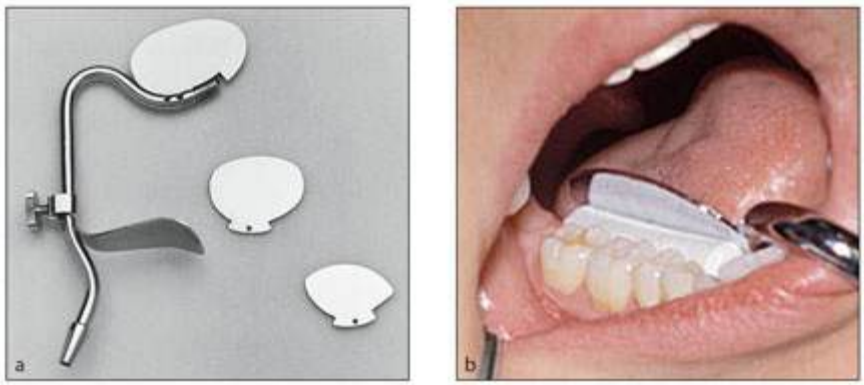


Fig 8-52 *(a)* The Svedopter tongue-retracting evacuation device is supplied with three sizes of vertical blades. *(b)* The Svedopter is used to hold the tongue away from the operating field.

Absorbent cotton rolls are placed adjacent to the Svedopter in the floor of the mouth and in the maxillary buccal vestibule adjacent to the opening of the parotid gland (Stensen) duct. The Svedopter is especially useful for preparation and cementation of fixed prostheses. It is less effective than the rubber dam for procedures in which total isolation from the fluids and vapors of the oral cavity is desired.

Hygoformic saliva ejector

The Hygoformic (Pulpdent) saliva ejector is used in the same way as the Svedopter for isolation in mandibular posterior areas, but it does not have a reflective blade (Fig 8-53). However, it is usually more comfortable and less traumatic to lingual tissues than is the Svedopter. For use, the saliva ejector must be re-formed (rebent) so that the evacuator tube passes under the chin, up over the incisal edges of the mandibular incisors, and then down to the floor of the mouth. The tongue-retracting coil should be loosened, or partially uncoiled, so that it extends posteriorly enough

to hold the tongue away from the operating field. The Hygoformic saliva ejector is also used with absorbent cotton for maximum effectiveness.

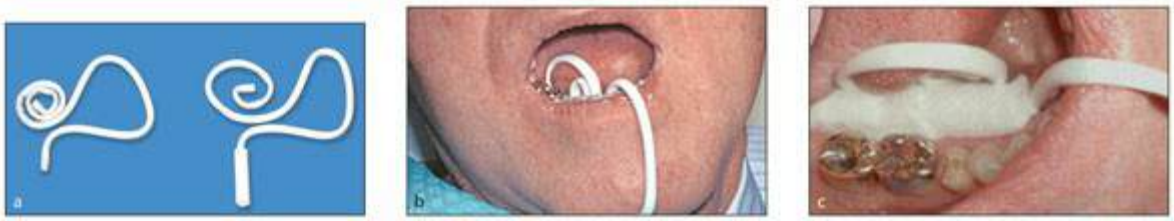


Fig 8-53 The Hygoformic saliva ejector should be routinely rebent to pass under the chin, over the incisal edges of the mandibular incisors, and then down to the floor of the mouth. The apparatus should usually be uncoiled slightly to extend further posteriorly. *(a, left)* Hygoformic saliva ejector as received; *(a, right)* Hygoformic saliva ejector that has been reshaped. *(b and c)* Isolation achieved with the Hygoformic saliva ejector.

Vac-Ejector

The Vac-Ejector Moisture Control System (Coltène/Whaledent) is made to facilitate isolation when restoring posterior teeth (Fig 8-54a). The Vac-Ejector incorporates a bite block, tongue retractor for mandibular areas, and high-speed suction attachment. It comes with three flexible deflectors, one universal deflector for operating on either side in the maxillary arch (Figs 8-54b and 8-54c) and one for each side when operating in mandibular areas (Fig 8-54d). The bite block is adjustable, by rotation, for large or small arches; in Fig 8-54, it is adjusted for large arches. Although this product appears complex to assemble correctly, operators soon become skilled at rapid assembly.



Fig 8-54 The Vac-Ejector provides a bite block, tongue retraction, and suction. *(a)* Parts of the Vac-Ejector; *(b)* assembled and ready for use for isolation of a maxillary posterior area; *(c)* in use for isolating the maxillary right posterior area; *(d)* in use for isolating the mandibular right posterior area.

Absorbent paper and cotton products

Absorbent materials are important in dentistry. Vacuum apparatuses remove fluids from the operating field by suctioning them; cotton and paper products help control fluids by absorbing them. Several types of absorbent cotton rolls are available in various diameters and lengths. These are placed into areas of the mouth where salivary gland ducts exit to absorb saliva and prevent salivary contamination of the operating field.

Isolation using absorbent materials with suctioning devices is less effective than using the rubber dam with suction, but in many procedures, the more complete isolation provided by the dam is unnecessary. In these situations, absorbent products are useful.

Small gauze sponges may be folded or rolled to substitute for cotton rolls. In addition, absorbent paper triangles, or parotid shields, such as Dri-Aid (Lorvic), are useful on the facial aspect of posterior teeth to absorb saliva secreted by the parotid gland (Fig 8-55).

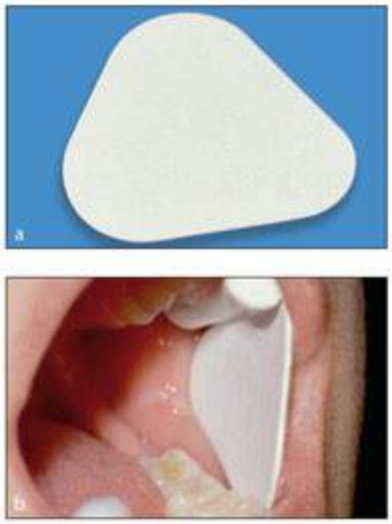


Fig 8-55*(a)* A parotid shield is triangular and made of absorbent paper. *(b)* A parotid shield may supplement a cotton roll in the buccal vestibule or may be used alone.

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